

Heat and Mass Transfer

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Syllabus

ME 1604 : HEAT AND MASS TRANSFER

Module I

Introduction to heat transfer – basic modes of heat transfer – conduction heat transfer – Fourier law of heat conduction – temperature dependence of thermal conductivity- general heat conduction equation in cartesian, cylindrical and spherical coordinates – boundary conditions – one-dimensional steady state conduction- critical insulation thickness -one-dimensional steady state conduction with heat generation -extended surface – two dimensional steady state heat conduction – conduction shape factor – unsteady state heat conduction in one dimension – lumped heat capacity system – semi-infinite solids with sudden change in surface temperature – Introduction to numerical methods in conduction.

Module II

Convective heat transfer – Newton's law of cooling – Prandtl number – laminar forced convection heat transfer from flat plates – fully developed laminar flow in pipes – turbulent forced convection – Reynolds' analogy – natural convection – natural convection heat transfer from vertical plates and horizontal tubes – condensation and boiling – film and drop wise condensation – film boiling and pool boiling – introduction to multiphase flow and heat transfer. Diffusion and convective mass transfer-Ficks law of diffusion.

Module III

Radiative transfer – electromagnetic radiation spectrum – thermal radiation – radiation properties- black body, gray body – monochromatic and total emissive power – Planck's law – Stefan-Boltzman law – Wien's displacement law – Kirchhoffs identity – shape factor- reciprocity relation – heat exchange between non black bodies; surface and shape resistances- electrical network analogy- heat transfer between parallel surfaces – radiation shields.

Module IV

Heat Exchangers: Type of heat exchangers- overall heat transfer coefficient -fouling factors -Logarithmic mean temperature difference (LMTD)- derivation of LMTD for parallel flow and counter flow heat exchangers-LMTD correction factor- effectiveness, NTU method of heat exchanger analysis- effectiveness derivation for parallel flow and counter flow heat exchangers. Design of parallel flow-counterflow-shell and tube multipass heat exchangers-condensers.

References:

- 1) Cengel, Heat Transfer Tata McGraw Hill
- 2) Holman J.P., "Heat Transfer", McGraw Hill International Students Edition
- 3) Incropera F.P. & De Witt D.P., "Fundamentals of Heat and Mass Transfer", John Wiley
- 4) Kreith F., "Heat Transfer", International Text Book Company
- 5) Gebhart B., "Heat Transfer", McGraw Hill
- 6) R.K. Rajput, "Heat and Mass Transfer", S Chand.
- 7) Venkanna, Fundamentals of HMT, PHI

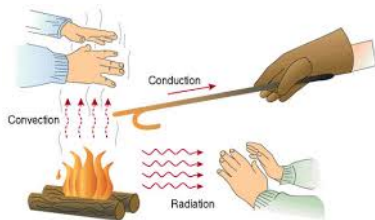
Data Book (1) Heat & Mass Transfer – C.P Kothandaraman, New Age International.

Data Book (2) Heat & Mass Transfer – Domkundwar, Dhanpat Rai.

Introduction

- 1 Thermodynamics $\implies$ Relation between heat and other forms of energy
- 2 Heat Transfer $\implies$ Rate of heat transfer in a system
- 3 Heat transfer rate form of Energy transfer (W)
 - 1 Conduction
 - 2 Convection
 - 3 Radiation
- 4 Potential $\implies$ Temperature difference (K)

Convection heat transfer



Convection

- 1 Convection- Heat transfer between solid surface and fluid
- 2 As a consequence of the motion of fluid relative to the surface
 - 1 Natural convection - Fluid motion resulting from the gravitational pull on buoyancy difference
 - 2 Forced convection - Fluid motion artificially created -by pump fan, blower

Convection heat transfer

Natural convection



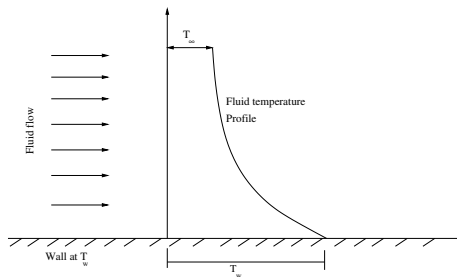
Forced convection



Convection heat transfer

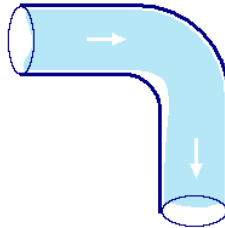
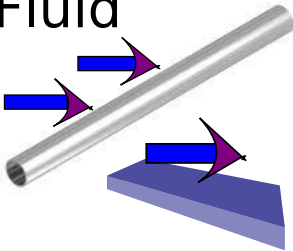
Newtons law of cooling

- 1 Convection heat transfer Q
- 2 $Q = hA(T_w - T_\infty)$
- 3 h - convection heat transfer coefficient (W/m^2K)
- 4 Same law for forced convection and natural convection
- 5 Task is to find h for different flow situations !!



Convection basics

Fluid



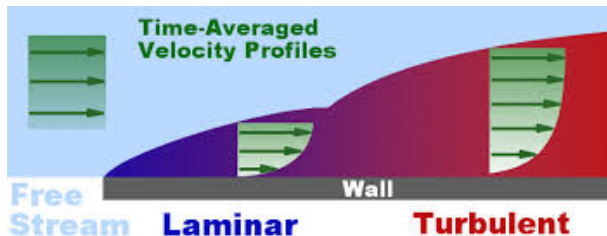
Some fluid mechanics

- 1 Identify the flow - External or Internal
- 2 Identify flow regime - Laminar ?, Turbulent ?

Basics

- 1 External flow Reynold number $\frac{\rho v x}{\mu}$
- 2 ρ - Density of the fluid $\frac{kg}{m^3}$, v -fluid velocity m/s, μ -dynamic viscosity $\frac{kgm}{s}$
 - 1 External flow 'x' for flow over flat plate, 'd' for flow across cylinder
 - 2 Critical Reynolds number external flow?
 - 3 Internal flow 'd'
 - 4 Critical Reynolds number internal flow?
 - 5 Hydraulic diameter $\frac{4A}{P}$

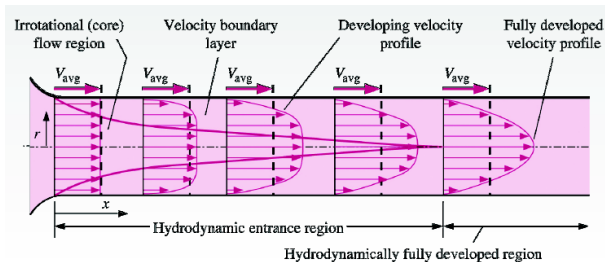
Hydrodynamic Boundary Layer



Hydrodynamic boundary layer

- 1 layer of fluid in the immediate vicinity of a bounding surface where the effects of viscosity are significant
- 2 at the edge of B.L velocity $0.99u_{\infty}$

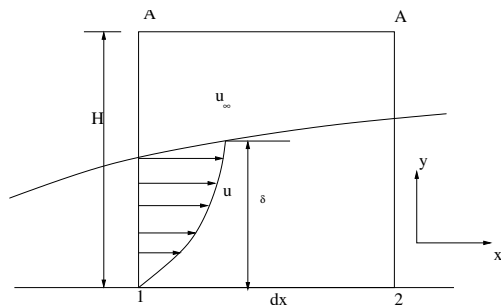
Boundary Layer Entry Length



Hydrodynamic boundary layer

- 1 Entry length (l_e) depends on Re
- 2 V_{avg} - constant ?
- 3 Fully developed flow after l_e

Hydrodynamic Boundary Layer



Hydrodynamic boundary layer

- 1 Why solving B.L for heat transfer problem ?
- 2 Assume $H > \delta$
- 3 Control volume 1, 2, A – A

Hydrodynamic Boundary Layer Analysis

1 The mass flow through plane 1 = $\int_0^H \rho u dy$

2 Momentum flow through plane 1 = $\int_0^H \rho u^2 dy$

3 Mass flow through plane 2 = $\int_0^H \rho u dy + \frac{d}{dx} \left(\int_0^H \rho u dy \right) dx$

4 Momentum flow plane 2 = $\int_0^H \rho u^2 dy + \frac{d}{dx} \left(\int_0^H \rho u^2 dy \right) dx$

5 mass flow through plane A-A = $\frac{d}{dx} \left(\int_0^H \rho u dy \right) dx$

6 momentum flow through plane A-A = $u_\infty \frac{d}{dx} \left(\int_0^H \rho u dy \right) dx$

Hydrodynamic Boundary Layer Analysis

► HBL

- 1 Net momentum flow out of C.V =

$$\frac{d}{dx} \left(\int_0^H \rho u^2 dy \right) dx - u_\infty \frac{d}{dx} \left(\int_0^H \rho u dy \right) dx \quad (1)$$

- 2 product formula $d(\eta\Phi) = \eta d\Phi + \Phi d\eta$

- 3 Therefore, $\eta d\Phi = d(\eta\Phi) - \Phi d\eta$, here $\Phi = \int_0^H \rho u dy$ and $\eta = u_\infty$

- 4 Consider $u_\infty \frac{d}{dx} \left(\int_0^H \rho u dy \right) dx =$

$$\frac{d}{dx} \left(u_\infty \int_0^H \rho u dy \right) dx - \frac{du_\infty}{dx} \left(\int_0^H \rho u dy \right) dx$$

$$1 \quad u_{\infty} \frac{d}{dx} \left(\int_0^H \rho u dy \right) dx =$$

$$\frac{d}{dx} \left(\int_0^H \rho u u_{\infty} dy \right) dx - \frac{du_{\infty}}{dx} \left(\int_0^H \rho u dy \right) dx$$

2 Substituting in Eq. (1), we get net momentum flow out of C.V =

$$\frac{d}{dx} \left(\int_0^H \rho u^2 dy \right) dx - \frac{d}{dx} \left(\int_0^H \rho u u_{\infty} dy \right) dx + \frac{du_{\infty}}{dx} \left(\int_0^H \rho u dy \right) dx$$

3 momentum flow out of C.V

$$= -\rho \frac{d}{dx} \int_0^H (u_{\infty} - u) u dy + \frac{du_{\infty}}{dx} \left(\int_0^H \rho u dy \right) dx \quad (2)$$

4 Force on plane 1 = pH plane 2 = $[p + \frac{dp}{dx}]H$, Net force
 $= -\frac{dp}{dx}H$

- 1 Shear force at the wall $-\tau_w dx = -\mu dx \frac{\partial u}{\partial y}|_{y=0}$
- 2 Substituting the net forces and momentum from Eq. (2)

$$-\tau_w - \frac{dp}{dx} H = -\rho \frac{d}{dx} \int_0^H (u_\infty - u) u dy + \frac{du_\infty}{dx} \left(\int_0^H \rho u dy \right) dx \quad (3)$$

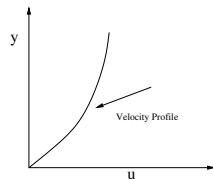
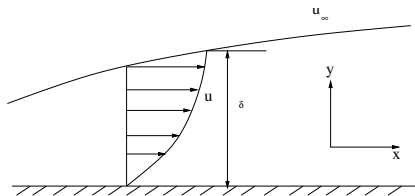
- 3 The Eq (3) is the integral momentum equation of the boundary layer
- 4 Consider the Bernoulli equation $\frac{p}{\rho} + \frac{u_\infty^2}{2} = \text{Constant}$
- 5 Differentiating $\frac{dp}{dx} \frac{1}{\rho} + u_\infty \frac{du_\infty}{dx} = 0$
- 6 $\frac{dp}{dx} = -\rho u_\infty \frac{du_\infty}{dx}$, Considering $\frac{dp}{dx} = 0 \implies \frac{du_\infty}{dx} = 0$

- 1 The momentum equation Eq (3) becomes

$$\mu dx \frac{\partial u}{\partial y} \Big|_{y=0} = \rho \frac{d}{dx} \int_0^H (u_\infty - u) u dy \quad (4)$$

- 2 The upper limit is changed to δ because $u = u_\infty$ for $y > \delta$
- 3 The Eq (4) becomes

$$\mu dx \frac{\partial u}{\partial y} \Big|_{y=0} = \rho \frac{d}{dx} \int_0^\delta (u_\infty - u) u dy \quad (5)$$



Velocity profile

- 1 Assume a velocity profile $u = C_1 + C_2y + C_3y^2 + C_4y^3$
- 2 Boundary conditions (i) $u = 0$ at $y = 0$ (ii) $u = u_{\infty}$ at $y = \delta$
- 3 (iii) $\frac{\partial u}{\partial y} = 0$ at $y = \delta$ (iv) $\frac{\partial^2 u}{\partial y^2} = 0$ at $y = 0$

- (i) $\Rightarrow C_1 = 0$, $\frac{\partial u}{\partial y} = C_2 + 2C_3y + 3C_4y^2$, $\frac{\partial^2 u}{\partial y^2} = 2C_3 + 6C_4y$
- (iv) $\Rightarrow C_3 = 0$, (ii) $\Rightarrow u_\infty = C_2\delta + C_4\delta^3$ (iii) $\Rightarrow C_2 + 3C_4\delta^2 = 0$
- Solving the two eqns we get, $C_2 = \frac{3u_\infty}{2\delta}$, $C_4 = -\frac{u_\infty}{2\delta^3}$
- The velocity profile equation

$$\frac{u}{u_\infty} = \frac{3y}{2\delta} - \frac{1}{2} \left(\frac{y}{\delta} \right)^3 \quad (6)$$

- The Eq (4) can be modified as

$$\mu dx \frac{\partial u}{\partial y} \Big|_{y=0} = \rho \frac{d}{dx} \int_0^\delta u_\infty \left(1 - \frac{u}{u_\infty} \right) u dy$$

- Substituting the velocity $\mu dx \frac{\partial u}{\partial y} \Big|_{y=0} =$

$$\frac{d}{dx} \left\{ \rho u_\infty^2 \int_0^\delta \left[\frac{3y}{2\delta} - \frac{1}{2} \left(\frac{y}{\delta} \right)^3 \right] \left[1 - \frac{3y}{2\delta} - \frac{1}{2} \left(\frac{y}{\delta} \right)^3 \right] dy \right\}$$

■ Integration gives $\frac{d}{dx} \left(\frac{39\rho u_{\infty}^2 \delta}{280} \right) = \frac{3\mu u_{\infty}}{2\delta}$

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- Separating variables

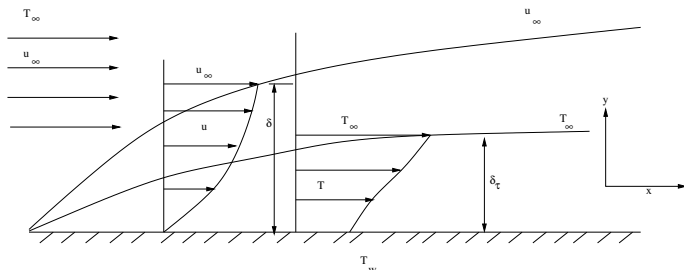
$$\delta d\delta = \frac{140\mu dx}{13\rho u_{\infty}} = \frac{140\nu dx}{13u_{\infty}} \quad (7)$$

- solving $\frac{\delta^2}{2} = \frac{140\nu x}{13u_{\infty}} + \text{const}$
- At $x = 0$, $\delta = 0$ therefore $\delta = 4.64 \sqrt{\frac{\nu x}{u_{\infty}}}$
- this can be modified as $\frac{\delta}{x} = \frac{4.64}{Re_x^{0.5}}$ where $Re_x = \frac{u_{\infty} x}{\nu}$

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- Exact solution $\frac{\delta}{x} = \frac{5}{Re_x^{0.5}}$



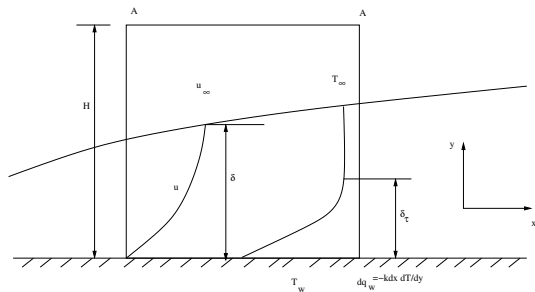
Prandtl number

- Thermal boundary layer - limit of temperature $T = 0.99 T_\infty$
- Prandtl number $Pr = \frac{\nu}{\alpha} = \frac{\mu c_p}{k}$, $Pr = \frac{\text{Viscous diffusion}}{\text{Thermal Diffusion}}$. Where μ - Dynamic viscosity of fluid in $\frac{\text{kgm}}{\text{s}}$
- c_p - specific heat of fluid $\frac{\text{kJ}}{\text{kg}^\circ\text{C}}$, k - conductivity of fluid $\frac{\text{kW}}{\text{ms}}$
- $Pr > 1$ Thermal boundary layer inside the HBL, $Pr = 1$ Both growing together $Pr < 1$ Thermal boundary layer thicker than HBL (liquid metals)

Thermal Boundary Layer

Control volume - boundary layer

- 1 $T = T_w$ at $y = 0$
- 2 $\frac{dT}{dy} = 0$, at $y = \delta_t$
- 3 $T = T_\infty$ at $y = \delta_t$
- 4 $\left(\frac{\partial^2 T}{\partial y^2}\right) = 0$ at $y = 0$
- 5 $\frac{\Theta}{\Theta_\infty} = \frac{T - T_w}{T_\infty - T_w} = \frac{3y}{2\delta_t} - \frac{1}{2} \left(\frac{y}{\delta_t}\right)^3$



$$1 \quad T = a_0 + a_1 y + a_2 y^2 + a_3 y^3$$

$$2 \quad \text{B.C (1)} \Rightarrow a_0 = T_w \Rightarrow T - T_w = a_1 y + a_2 y^2 + a_3 y^3$$

$$3 \quad \text{B.C (2)} \Rightarrow a_1 + 2a_2 \delta_t + 3a_3 \delta_t^2 = 0$$

$$4 \quad \text{B.C (3)} \Rightarrow a_1 \delta_t + a_2 \delta_t^2 + a_3 \delta_t^3 = T_\infty - T_w$$

$$5 \quad \text{B.C (4)} \Rightarrow 2a_2 + 6a_3 y = 0|_{(y=0)} \Rightarrow a_2 = 0$$

$$6 \quad a_1 + 3a_3 \delta_t^2 = 0 \Rightarrow a_1 = -3a_3 \delta_t^2$$

$$7 \quad a_1 \delta_t + a_3 \delta_t^3 = T_\infty - T_w; -3a_3 \delta_t^3 + a_3 \delta_t^3 = T_\infty - T_w$$

$$8 \quad \Rightarrow a_3 = \frac{-(T_\infty - T_w)}{2\delta_t^3}, a_1 = \frac{3(T_\infty - T_w)}{2\delta_t}$$

$$9 \quad T - T_w = \frac{3(T_\infty - T_w)y}{2\delta_t} - \frac{(T_\infty - T_w)y^3}{2\delta_t^3}$$

$$10 \quad \frac{\Theta}{\Theta_\infty} = \frac{T - T_w}{T_\infty - T_w} = \frac{3y}{2\delta_t} - \frac{1}{2} \left(\frac{y}{\delta_t} \right)^3$$

- 1 Energy balance of the control volume
- 2 Energy convected in + viscous work within the element + heat transfer at the wall = energy convected out.

- 3 Energy convected in at 1-A = $\rho c_p \int_0^H u T dy$

- 4 Energy convected out at 2-A = $\rho c_p \int_0^H u T dy + \frac{d}{dx} \left(\rho c_p \int_0^H u T dy \right) dx$

- 5 Mass flow through A-A = $\frac{d}{dx} \left(\int_0^H \rho u dy \right) dx$

- 6 Energy carried through A-A = $c_p T_\infty \frac{d}{dx} \left(\int_0^H \rho u dy \right) dx$

- 7 Viscous shear force = $\mu \frac{\partial u}{\partial y} dx$

- 8 Distance moved through unit time = $\frac{\partial u}{\partial y} dy$

$$1 \quad \text{Viscous energy delivered} = \mu \left[\int_0^H \left(\frac{du}{dy} \right)^2 dy \right] dx$$

$$2 \quad \text{Heat transfer } dq_w = -k dx \frac{\partial T}{\partial y} \Big|_w$$

$$3 \quad \text{Energy balance} =$$

$$\frac{d}{dx} \left(\int_0^H (T_\infty - T) u dy \right) + \frac{\mu}{\rho c_p} \left[\int_0^H \left(\frac{du}{dy} \right)^2 dy \right] = \alpha \frac{\partial T}{\partial y} \Big|_w$$

$$4 \quad \frac{d}{dx} \left(\int_0^H (T_\infty - T) u dy \right) = \frac{d}{dx} \left(\int_0^H (\theta_\infty - \theta) u dy \right)$$

$$5 \quad = \theta_\infty u_\infty \frac{d}{dx} \left(\int_0^H \left[1 - \frac{3y}{2\delta_t} + \frac{1}{2} \left(\frac{y}{\delta_t} \right)^3 \right] \left[\frac{3y}{2\delta} - \frac{1}{2} \left(\frac{y}{\delta} \right)^3 \right] dy \right) =$$

$$\alpha \frac{\partial T}{\partial y} \Big|_{y=0} = \frac{3\alpha\theta_\infty}{2\delta_t}$$

- 1 Assume thermal boundary layer is thinner than hydrodynamic boundary layer
- 2 integrating to $y = \delta_t$ since integrand is zero for $y > \delta_t$, also take $\zeta = \frac{\delta_t}{\delta}$
- 3 $\theta_\infty u_\infty \frac{d}{dx} \left[\delta \left(\frac{3\zeta^2}{20} - \frac{3\zeta^4}{280} \right) \right] = \frac{3\alpha\theta_\infty}{2\zeta\delta}$
- 4 $\delta > \delta_t, \zeta < 1$ neglecting terms involving ζ^4
- 5 $\frac{3}{20} \theta_\infty u_\infty \frac{d(\delta\zeta^2)}{dx} = \frac{3\alpha\theta_\infty}{2\zeta\delta}$
- 6 $\frac{1}{10} u_\infty \left(2\delta\zeta \frac{d\zeta}{dx} + \zeta^2 \frac{d\delta}{dx} \right) = \frac{\alpha}{\delta\zeta}$
- 7 $\frac{1}{10} u_\infty \left(2\delta^2\zeta^2 \frac{d\zeta}{dx} + \zeta^3 \delta \frac{d\delta}{dx} \right) = \alpha$
- 8 we have; from Eq. (7) $\delta d\delta = \frac{140\nu dx}{13u_\infty}$ and $\delta^2 = \frac{280\nu x}{13u_\infty}$

- 1 simplifying; $\zeta^3 + 4x\zeta^2 \frac{d\zeta}{dx} = \frac{13\alpha}{14\nu}$

- 2 Since $\zeta^2 \frac{d\zeta}{dx} = \frac{1}{3} \frac{d\zeta^3}{dx}$

- 3 $\zeta^3 + \frac{4}{3}x \frac{d\zeta^3}{dx} = \frac{13\alpha}{14\nu}$

- 4 The solution to this equation is $\zeta^3 = Cx^{\frac{-3}{4}} + \frac{13\alpha}{14\nu}$

- 5 Boundary conditions $\delta_t = 0$ at $x = x_0$ and $\zeta = 0$ at $x = x_0$

- 6 Final solution is $\zeta = \frac{\delta_t}{\delta} = \frac{1}{1.026} Pr^{-\frac{1}{3}} \left[1 - \left(\frac{x_0}{x} \right)^{\frac{3}{4}} \right]^{\frac{1}{3}}$

- 7 $h = \frac{-k \frac{\partial T}{\partial y}|_w}{T_w - T_\infty} = \frac{3k}{2\delta_t} = \frac{3k}{2\zeta\delta}$

- 8 Substituting $\delta = \frac{4.64x}{Re_x^{\frac{1}{2}}}$ and ζ

- 9 $h_x = 0.332k \left(\frac{u_\infty}{\nu x} \right)^{\frac{1}{2}} Pr^{\frac{1}{3}} \left[1 - \left(\frac{x_0}{x} \right)^{\frac{3}{4}} \right]^{-\frac{1}{3}}$

- 10 $Nu_x = 0.332 Re_x^{\frac{1}{2}} Pr^{\frac{1}{3}} \left[1 - \left(\frac{x_0}{x} \right)^{\frac{3}{4}} \right]^{-\frac{1}{3}}$

- 11 Nusslet number = $\frac{\text{Convective heat transfer}}{\text{Conductive heat transfer}} \quad Nu = \frac{hx}{k} \text{ or } Nu = \frac{hD}{k}$

- We have $h_x = 0.332k \left(\frac{u_\infty}{\nu x} \right)^{\frac{1}{2}} Pr^{\frac{1}{3}} \left[1 - \left(\frac{x_0}{x} \right)^{\frac{3}{4}} \right]^{-\frac{1}{3}}$

- If $x_0 = 0 \Rightarrow h_x = 0.332k \left(\frac{u_\infty x}{\nu} \right)^{\frac{1}{2}} Pr^{\frac{1}{3}}$

- Average heat transfer coefficient $h_{av} = \frac{\int_0^l h_x dx}{\int_0^l dx}$

- $h_{av} = \frac{\int_0^l 0.332k \left(\frac{u_\infty}{\nu x} \right)^{\frac{1}{2}} Pr^{\frac{1}{3}} dx}{\int_0^l dx}$

- $h_{av} = \frac{2 \times 0.332k \left(\frac{u_\infty l}{\nu} \right)^{\frac{1}{2}} Pr^{\frac{1}{3}}}{l}$

- $h_{av} = 2 \times 0.332k \left(\frac{u_\infty}{\nu l} \right)^{\frac{1}{2}} Pr^{\frac{1}{3}}$

- $h_{av}|_{x=l} = 2 \times h_x|_{x=l}$

- Properties of the fluid at mean film temperature $T_f = \frac{T_w + T_\infty}{2}$

Other Relations

- Constant heat flux surface $Nu_x = \frac{hx}{k} = 0.453 Re_x^{\frac{1}{2}} Pr^{\frac{1}{3}}$
- Churchill and Ozoe for $Re_x Pr > 100$ condition

$$Nu_x = \frac{0.3387 Re_x^{\frac{1}{2}} Pr^{\frac{1}{3}}}{\left[1 + \left(\frac{0.0468}{Pr} \right)^{\frac{2}{3}} \right]^{\frac{1}{4}}}$$

Reynold-Colburn analogy

- The shear stress at the wall $\tau_w = C_f \frac{\rho u_\infty^2}{2}$
- Also we have $\tau_w = \mu \frac{\partial u}{\partial y}|_w$
- From the velocity profile equation Eq. (6) we have
- $\tau_w = \frac{3}{2} \frac{\mu u_\infty}{\delta}$
- Substituting for the boundary layer thickness
- $\tau_w = \frac{3\mu u_\infty}{2 \times 4.64} \left(\frac{u_\infty}{\nu x} \right)^{\frac{1}{2}}$
- Combining the equations $\frac{C_{fx}}{2} = \frac{3\mu u_\infty}{2 \times 4.64} \left(\frac{u_\infty}{\nu x} \right)^{\frac{1}{2}} \frac{1}{\rho u_\infty^2}$
- Gives $\frac{C_{fx}}{2} = 0.323 Re_x^{-\frac{1}{2}}$
- Considering the exact solution we get $\frac{C_{fx}}{2} = 0.332 Re_x^{-\frac{1}{2}}$

- Consider, $\frac{Nu_x}{Re_x Pr} = \frac{h_x}{\rho c_p u_\infty} = 0.332 Re_x^{-\frac{1}{2}} Pr^{-\frac{2}{3}}$
- The group Stanton number $St_x = \frac{h_x}{\rho c_p u_\infty}$
- Therefore $St_x Pr^{\frac{2}{3}} = 0.332 Re_x^{-\frac{1}{2}}$
- Thus $St_x Pr^{\frac{2}{3}} = \frac{C_{fx}}{2}$ - This is the Reynolds -Colburn Analogy

Laminar Flow -Flow Inside ducts HBL and TBL developed

- For Circular tubes Constant T $Nu = 3.657$ and For const. heat flux - $Nu = 4.364$
- For Square tube Constant T $Nu = 2.98$ and For const. heat flux - $Nu = 3.6$

Turbulent Flow -Flow over flat plate

- For $5 \times 10^5 < Re_x < 10^7$ Use $St_x Pr^{\frac{2}{3}} = 0.0296 Re_x^{-0.2}$
- For $10^7 < Re_x < 10^9$ Use $St_x Pr^{\frac{2}{3}} = 0.185 (\log Re_x)^{-2.584}$

Empirical Relations

Flow inside tubes

- Properties at average fluid bulk temperature

$$T_b = (T_{b1} + T_{b2})/2$$

- In fully developed turbulent flow in smooth tubes

$Nu = 0.023Re^{0.8}Pr^n$ [$n=0.4$ for heating of fluid, $n=0.3$ for cooling the fluid]- Properties at average fluid bulk temperature. - Dittus Boelter relation.

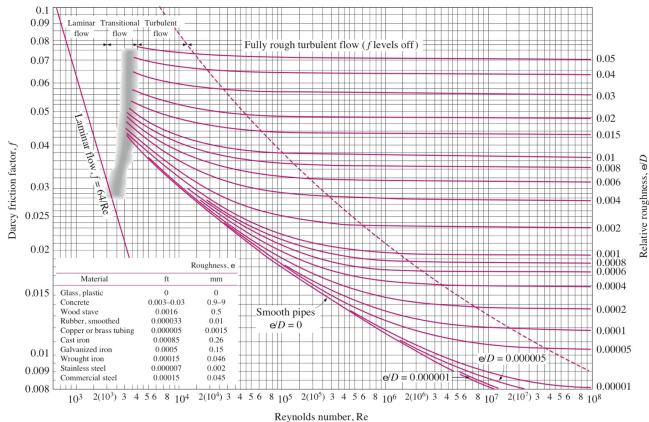
- Flow inside tube - turbulent flow

$Nu = 0.027Re^{0.8}Pr^{\frac{1}{3}} \left(\frac{\mu}{\mu_w} \right)^{0.14}$ Sieder and Tate Equation μ_w
Dynamic viscosity at wall temperature

- Petukhov equation for fully developed turbulent flow inside

tubes $Nu = \frac{(f/8)RePr}{1.07 + 12.7(f/8)^{\frac{1}{2}}(Pr^{\frac{2}{3}} - 1)} \left(\frac{\mu}{\mu_w} \right)^n$, $n=0.11$ for
 $T_w > T_b$ and $n=0.25$ for $T_w < T_b$ find f from graph

Moody diagram



- From Reynolds number and relative surface roughness friction factor can be found

■ Cross flow over circular cylinder

$$Nu = \frac{hD}{k} = C \left(\frac{u_{\infty} D}{\nu} \right)^n Pr^{\frac{1}{3}} \quad (8)$$

Re	C	n
0.4-4	0.989	0.33
4-40	0.911	0.385
40-4000	0.683	0.466
4000-40000	0.193	0.618
40000-400000	0.0266	0.805

- Heat transfer from tubes in cross flow (Eckert and Drake)

$$Nu = (0.43 + 0.50Re^{0.5})Pr^{0.38} \left(\frac{Pr}{Pr_w} \right)^{0.25} \text{ for } 1 < Re < 10^3$$

- $Nu = 0.25Re^{0.6})Pr^{0.38} \left(\frac{Pr}{Pr_w} \right)^{0.25}$ for $10^3 < Re < 2 \times 10^5$

- Heat transfer from tubes in cross flow Churchill and Bernstein

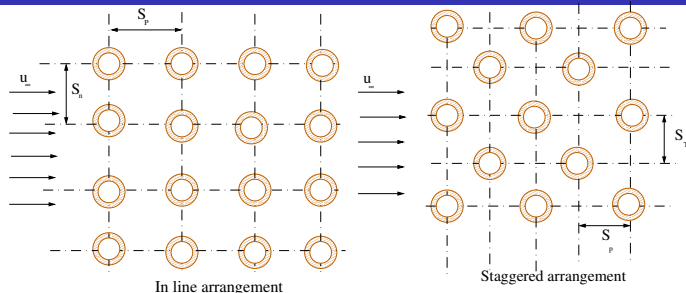
$$Nu = 0.3 + \frac{0.62Re^{\frac{1}{2}}Pr^{\frac{1}{3}}}{[1+(0.4/Pr)^{\frac{2}{3}}]^{\frac{1}{4}}} \left[1 + \left(\frac{Re}{282000} \right)^{\frac{5}{8}} \right]^{\frac{4}{5}} \text{ for } 10^2 < Re < 10^7$$

Spheres

- Whitaker developed the following equation for gases and liquids flowing past spheres

$$Nu = 2 + (0.4Re^{\frac{1}{2}} + 0.06Re^{\frac{2}{3}})Pr^{0.4}(\mu_{\infty}/\mu_w)^{\frac{1}{4}} \text{ valid for } 3.5 < Re < 8 \times 10^4$$

Bank of Tubes

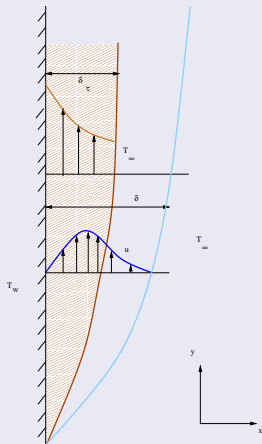


Tube banks

- Maximum velocity in line arrangement $u_{max} = u_\infty [S_n / (S_n - d)]$
- Maximum velocity in staggered arrangement $u_{max} = \frac{u_\infty (S_n/2)}{[(S_n/2)^2 + S_p^2]^{1/2} - d}$
- Use Eq- (8) and the correction factor for tube banks of 10 rows or more.

Natural convection

Boundary layer



Natural convection

- Natural convection free convection
- Movement of fluid caused due to density changes arising from heating or cooling
- Heat transfer $Q = hA(T_w - T_\infty)$

Boundary layer analysis -Natural convection vertical plate

- Momentum flux through control volume dx-dy

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} - \rho g + \mu \frac{\partial^2 u}{\partial y^2}$$

- Pressure gradient $\frac{\partial p}{\partial x} = -\rho_{\infty} g$

- Substituting, $\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = g(\rho_{\infty} - \rho) + \mu \frac{\partial^2 u}{\partial y^2}$

- we have , $\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p = \frac{1}{V_{\infty}} \frac{V - V_{\infty}}{T - T_{\infty}} = \frac{\rho_{\infty} - \rho}{\rho(T - T_{\infty})}$

- Substituting $\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = g\rho\beta(T - T_{\infty}) + \mu \frac{\partial^2 u}{\partial y^2}$ -
Equation of motion for free - convection boundary layer

- Integral momentum equation $\frac{d}{dx} \left(\int_0^{\delta} \rho u^2 dy \right) =$

$$-\tau_w + \int_0^{\delta} \rho g \beta (T - T_{\infty}) dy = -\mu \frac{\partial u}{\partial y} \Big|_{y=0} + \int_0^{\delta} \rho g \beta (T - T_{\infty}) dy$$

- Boundary conditions $T = T_w$ at $y = 0$, $T = T_\infty$ at $y = \delta$, $\frac{\partial T}{\partial y} = 0$ at $y = \delta$
- Temperature profile $\frac{T - T_\infty}{T_w - T_\infty} = \left(1 - \frac{y}{\delta}\right)^2$
- Boundary conditions $u = 0$ at $y = 0$, $u = 0$ at $y = \delta$, $\frac{\partial u}{\partial y} = 0$ at $y = \delta$
- additional boundary condition $\frac{u}{u_x} = \frac{\beta \delta^2 g (T_w - T_\infty) y}{4 u_x \nu \delta} \left(1 - \frac{y}{\delta}\right)^2$
- $\frac{\delta}{x} = 3.93 Pr^{-1/2} (0.952 + Pr)^{1/4} Gr_x^{-1/4}$
- Grashof number $Gr_x = \frac{g \beta (T_w - T_\infty) x^3}{\nu^2}$, where $\beta = 1/T_f$ and $T_f = (T_w + T_\infty)/2$ in absolute kelvin (K)
- $Nu_x = 0.508 Pr^{1/2} (0.952 + Pr)^{-1/4} Gr_x^{1/4}$
- Average heat transfer coefficient $h_{av} = \frac{4}{3} h_{x=L}$

Empirical relations for free convection

- Flow is considered to be Turbulent for $Gr > 10^8 - 10^9$
- Free convection Nusslet number $Nu = C(GrPr)^m$, The product $GrPr = Ra$, The Rayleigh number
- Properties of fluid @ $T_f = (T_\infty + T_w)/2$

Table : Isothermal surfaces

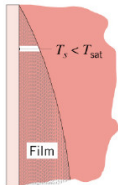
Geometry	GrPr	C	m
Vertical planes and cylinders	$10^4 - 10^9$	0.59	1/4
	$10^9 - 10^{13}$	0.1	1/3
Horizontal cylinders	$10^4 - 10^9$	0.53	1/4
	$10^9 - 10^{12}$	0.13	1/3
Upper surface of heated plates or lower surface of cooled plates	$2 \times 10^4 - 8 \times 10^6$	0.54	1/4
	$8 \times 10^6 - 10^{11}$	0.15	1/3
Lower surface of heated plates	$10^5 - 10^{11}$	0.27	1/4

Empirical relations for free convection vertical surfaces

- Constant heat flux modified $Gr_x^* = Gr_x Nu_x = \frac{g\beta q_w x^4}{k\nu^2}$
- Laminar range $Nu_x = hx/k = 0.6(Gr_x^* Pr)^{1/5}$ for $10^5 < Gr_x^* < 10^{11}$, $q_w = \text{constant}$
- Turbulent range $Nu_x = hx/k = 0.17(Gr_x^* Pr)^{1/4}$ for $2 \times 10^{13} < Gr_x^* Pr < 10^{16}$, $q_w = \text{constant}$
- Average heat transfer coefficient $h = 5/4 h_{x=L}$
- Simplified equation for air
Vertical plane or cylinder for $10^4 < GrPr < 10^9$,
 $h = 1.42 \left(\frac{\Delta T}{L} \right)^{1/4}$
- for $GrPr > 10^9$, $h = 1.31(\Delta T)^{1/3}$, $\Delta T = T_w - T_\infty$,
 $L = \text{Vertical or horizontal dimension, m}$

Condensation

Condensation occurs when the surface temperature is less than the saturation temperature of an adjoining vapor.

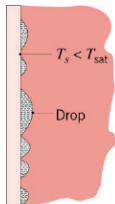


Filmwise condensation

Film Condensation

- Film Condensation-Entire surface is covered by the condensate
- Flows continuously from the surface
- Provides a resistance to heat transfer between the vapor and the surface.
- Thermal resistance is reduced through use of short vertical surfaces and horizontal cylinders.

Dropwise Condensation



Drop wise condensation

Dropwise Condensation

- Dropwise Condensation - Surface is covered by drops ranging from a few micrometers to agglomerations
- Thermal resistance is greatly reduced due to absence of a continuous film.
- Surface coatings may be applied to inhibit wetting and stimulate dropwise condensation.

- Heat transfer coefficient for film condensation on a vertical plate

- Local heat transfer coefficient $h_x = \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}k_l^3}{4\mu_l(T_{sat} - T_s)x} \right]^{1/4}$

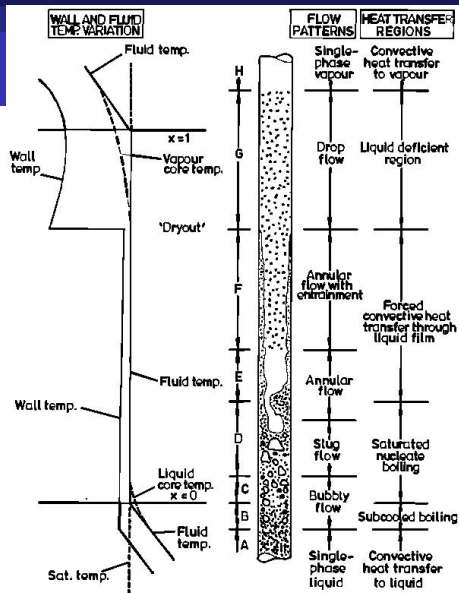
- Average heat transfer coefficient $h_{av} = 4/3 h_{x=L}$

- Average heat transfer coefficient

$$h_{av} = 0.943 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}k_l^3}{\mu_l(T_{sat} - T_s)L} \right]^{1/4}$$

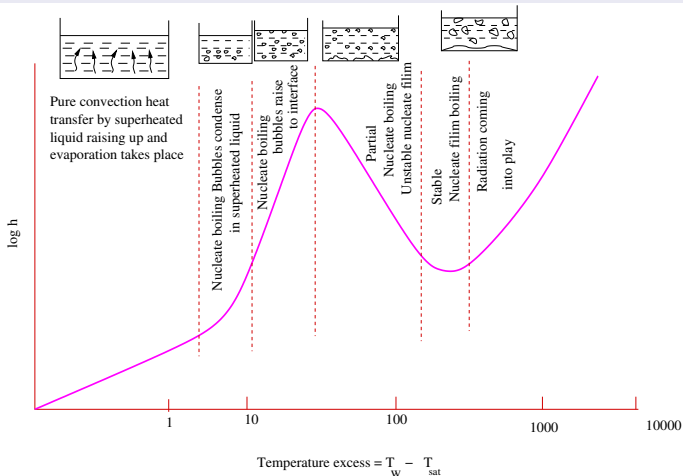
Boiling

- Boiling occurs at the solid liquid interface
- Liquid is at saturation temperature T_{sat}
- Surface maintained at a temperature above the saturation temperature T_{sat} of the liquid
- Boiling is called pool boiling in the absence of bulk fluid flow
- Flow boiling (or forced convection boiling) in the presence of bulk fluid flow



Boiling

Typical boiling curve for water



Diffusion and Mass Transfer

- Mass transfer associated with convection and concentration gradient
- Ficks Law of Diffusion -mass flux proportional to concentration gradient
- $\frac{m_A}{A} = -D \frac{\partial C_A}{\partial x}$, D= Diffusion coefficient, m^2/s , m_A - mass flux per unit time , kg/s, C_A - mass concentration of component A per unit volume, kg/m^3
- Diffusive mass flux of component A, kg/s
 $m_A = KA(C_{A1} - C_{A2})$, K = mass transfer coefficient, m/s, C_{A1} , C_{A2} = concentrations through which diffusion occurs, kg/m^3
- Sherwood number $\frac{Kd}{D} = f(Re_x, Sc)$, Schmidt number $Sc = \frac{\nu}{D}$
- Gilliland equation for vaporization of liquid into air inside circular column- $\frac{Kd}{D} = 0.023 \left[\frac{\rho u_m d}{\mu} \right]^{0.83} \left[\frac{\nu}{D} \right]^{0.44}$